

Saikumar Chedalu

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Summary

Software Engineer with around 8 years of experience building scalable backend systems, distributed applications, and cloud-native services with a strong focus on **Python**, **AWS**, and modern AI technologies. Experienced in developing **LLM-powered applications**, **RAG pipelines**, **agentic AI workflows**, and intelligent automation systems using **AWS Bedrock**, **LangChain**, and **LangGraph**. Strong background in distributed systems, API development, cloud infrastructure, data processing, and production troubleshooting. Proven ability to design, build, deploy, and support end-to-end AI applications that combine large language models, semantic retrieval, and cloud-native architecture.

Technology Stack & Skills

Languages: Python, SQL, JavaScript, TypeScript

AI / LLM: OpenAI API, Prompt Engineering, LLM Integration, LangChain, Vector Search

Frameworks: FastAPI, Flask, Node.js (Basic), React

Cloud: AWS (EC2, Lambda, S3, CloudWatch, DynamoDB)

Data Processing: Pandas, NumPy, Log Analysis, Data Pipelines

Architecture: Distributed Systems, Microservices, REST APIs, Event-Driven Systems

Messaging: Apache Kafka

Databases: PostgreSQL, MySQL

DevOps: Docker, CI/CD, Git

Systems & Networking: TCP/IP, DNS, HTTP, TLS

Work Experience

Amazon – Software Development Engineer, Virginia

Dec 2024 - Current

- Developed **RAG-based AI services** using **AWS Bedrock**, **LangChain**, and semantic retrieval workflows to improve contextual response quality and knowledge retrieval capabilities for conversational AI systems.
- Built **FastAPI-based microservices** integrating foundation models, backend AWS services, and distributed processing components to support intelligent workflow automation and low-latency AI interactions.
- Designed document ingestion and retrieval pipelines involving document chunking, embeddings generation, vector indexing, and semantic search workflows for enterprise knowledge discovery.
- Developed **agentic AI workflows** capable of multi-step reasoning, tool execution, context management, and workflow orchestration using **LangGraph** and LLM-powered planning techniques.
- Integrated **Claude** and Bedrock-hosted foundation models into distributed backend services supporting intelligent customer interactions and AI-powered automation workflows.
- Built internal tooling for prompt evaluation, response quality analysis, latency measurement, workflow observability, and end-to-end tracing of AI-driven service requests.
- Implemented conversation memory management, context enrichment strategies, and prompt orchestration techniques to improve long-running AI interactions and response accuracy.

- Worked with **SageMaker-based evaluation workflows** to benchmark model performance, compare prompt strategies, and analyzing inference behavior across multiple LLM-powered use cases.
- Developed observability solutions using **CloudWatch**, structured logging, and tracing frameworks to troubleshoot AI workflow failures and monitor production inference systems.
- Collaborated with applied scientists, product managers, and platform engineers to design scalable AI architecture, produce LLM-powered features, and improve operational excellence for customer-facing AI applications.

Tools: Python, FastAPI, AWS Bedrock, SageMaker, OpenSearch, LangChain, LangGraph, Kafka, S3, Lambda, DynamoDB, CloudWatch, Docker.

State Street – Full Stack Developer, New Jersey

Oct 2022 – Dec 2024

- Developed **Python-based backend services** supporting data-intensive financial applications, reporting platforms, and enterprise analytics workflows.
- Built data processing pipelines using **Python**, SQL, and Pandas to transform, validate, and analyze large-scale structured datasets.
- Developed AI-assisted reporting solutions using **LangChain** and LLM APIs to automate financial data summarization, insight generation, and reporting workflows.
- Built semantic search capabilities for internal datasets using embeddings-based retrieval techniques and intelligent document search workflows.
- Developed **FastAPI services** exposing AI-powered insights, data analytics workflows, and backend processing capabilities to internal applications.
- Implemented real-time processing workflows using **Kafka**, enabling event-driven data movement and continuous processing across enterprise systems.
- Designed and optimized complex SQL queries supporting large-scale analytics, reporting, and operational processing requirements.
- Built and deployed cloud-hosted backend applications within Azure environments supporting scalable processing and storage workflows.
- Worked extensively with financial datasets performing validation, transformation, aggregation, and operational analysis for business-critical applications.
- Collaborated with engineering and analytics teams to scope investigations, analyze trends, and build scalable solutions supporting data-driven decision-making.

Tools: Python, SQL, Kafka, Azure, LangChain, FastAPI, PostgreSQL, Docker

Sonata Software - Software Developer, Bengaluru

Jan 2019 - Aug 2022

- Developed backend services using **Python** and **FastAPI** supporting enterprise operational workflows, business process automation, and application integrations.
- Built REST APIs and backend processing components responsible for handling high-volume business transactions and distributed application communication.
- Developed SQL-based data transformation, validation, and aggregation workflows supporting reporting systems and operational analytics requirements.
- Implemented event-driven processing pipelines using **Kafka** to enable asynchronous communication and real-time data movement between enterprise systems.
- Built data processing utilities for structured business datasets, improving automation and reducing manual operational effort across applications.

- Worked with AWS-hosted environments and Docker-based deployments supporting scalable application delivery and operational reliability.
 - Participated in production troubleshooting, backend debugging, root cause analysis, and application performance improvement initiatives.
 - Contributed to CI/CD workflows, release automation, deployment support, and operational monitoring activities across multiple enterprise applications.
 - Collaborated with cross-functional teams during application design discussions, implementation planning, testing activities, and production support operations.
- Tools: Python, Java, PostgreSQL, Apache Kafka, REST APIs, Docker, Git

Tools: Python, FastAPI, SQL, AWS, Kafka, Docker, Git

Education

Auburn University at Montgomery, Alabama

Master of Science, Information Systems

Avanthi College, Hyderabad

Bachelor of Science, Computer Science